

How to Clean a Clean Room

A clean room is a controlled environment where a particle 200 times smaller than a human hair can ruin the product. Cleaning one is a discipline of its own.

What a Clean Room Is

A clean room controls the concentration of airborne particles to a specified limit. Typical office air carries 500,000 to 1,000,000 particles (0.5 micron or larger) per cubic foot. A Class 100 clean room allows no more than 100. Air flow, pressure, temperature, humidity, and HEPA filtration are all engineered; the cleaning program is part of that engineering, not an add on.

CLASS	MAX PARTICLES PER CUBIC FOOT (0.5 MICRON+)	TYPICAL USE
Class 100	100	Semiconductor, sterile pharma
Class 1,000	1,000	Pharma, medical device
Class 10,000	10,000	Electronics, biotech support

For perspective: a person standing still sheds about 100,000 particles per minute. Walking at a normal pace, about 5,000,000. That is why conduct rules matter as much as chemicals.

Where Contamination Comes From

- **People:** skin flakes, hair, oils, cosmetics, perfume, and clothing lint. People are the number one source.
- **Facilities:** paint, construction dust, HVAC debris, and spills.
- **Tools:** wear particles, lubricants, and the wrong cleaning tools (brooms, cotton mops, and dusters generate particles instead of removing them).
- **Fluids and product:** chemical residue, floor finish, and process debris.

Gowning & Conduct

Follow the site's gowning procedure exactly. Gloves, face mask, and head cover are standard nearly everywhere; smocks or full suits with boot covers are required in higher classes. Then the conduct rules:

- ☐ Personal items (keys, phones, jewelry, cigarettes) stay in the locker outside the gowning room
- ☐ No eating, gum, or cosmetics of any kind inside
- ☐ Only approved clean room paper, pens, and wipers come in
- ☐ Walk at a normal pace; no fast movement, leaning on surfaces, or horseplay

- ☐ Gowns never leave the clean room; torn or soiled garments get replaced
- ☐ Nobody enters while ill, especially with respiratory symptoms

Approved Supplies Only

Every mop, wiper, vacuum, and chemical must be approved for the room's class and on the client's approved list. Standard janitorial supplies never cross the line. Typical clean room kit: approved cleaning and disinfecting solutions (often with deionized water), clean room mops and buckets, clean room wipers, and a HEPA vacuum if the site allows one.

Typical Cleaning Tasks & Frequencies

Every clean room runs on a site specific schedule agreed with the client. These patterns are common:

TASK	TYPICAL FREQUENCY
Change tacky mats at entrances	Every 2 hours
Wet mop floors with approved cleaner & DI water	1 to 2 times per shift
Remove trash (gloved; waste containers never enter the room)	Once per shift
Wipe work surfaces with approved wipers	Per shift to weekly, by zone
Wipe walls, doors, trim & pass throughs	Weekly
Vacuum floors (HEPA, if allowed)	Weekly to monthly

THE GOLDEN RULE

The client's local procedures always supersede general guidance. Before the first shift, walk the program with clean room management: what counts as clean, how it is measured, which materials are approved, and when cleaning is allowed.